GATE

Computer Science & Information Technology

Computer Organization & Architecture

Instruction & Addressing Modes

DPP: 2

Q1 A relative branch mode type instruction is stored in memory starting from address 240. The branch is made to an address 140. What should be the value of relative address field of the instruction, if each instruction is stored on 2 memory locations?
Note: All numbers are in decimal

Q2 Consider the following:
1. Operation code
2. Source operand reference
3. Result operand reference
4. Next instruction reference

Which of the above are typical elements of machine instructions?
(A) 1, 2 and 3 only
(B) 1, 2 and 4 only
(C) 3 and 4 only
(D) 1, 2, 3 and 4

Q3 Which addressing mode helps to access table data in memory efficiently?
(A) Indirect mode
(B) Immediate mode
(C) Auto-increment or Auto-decrement mode
(D) Index mode

Q4 An addressing mode in which the location of the data is contained within the mnemonic, is known as?
(A) Immediate addressing mode
(B) Implied addressing mode
(C) Register addressing mode
(D) Direct addressing mode

Q5 The addressing modes used for source operand in the following instructions are respectively?
$R1 \leftarrow \#5$
$R1 \leftarrow M[5000]$
$R1 \leftarrow M[R2]$
(A) Implied, direct, register
(B) Implied, direct, register indirect
(C) Immediate, direct, register indirect
(D) Immediate, direct, register

Q6 Consider a PC-relative mode type branch instruction which takes branch on address 720 in memory. The instruction has offset value 160. What is the starting address of this instruction in memory, if each instruction is stored in memory on 4 locations?
Note: All numbers are in decimal

Q7 Consider the system in which in fetch cycle complete instruction is fetched. Which of the following addressing modes do(es) not require memory access for operand after fetch cycle?
(A) Register Mode
(B) Register Indirect Mode
(C) Indirect Mode
(D) Indexed Mode

Answer Key

Q1	-102~-102
Q2	(A)
Q3	(C)
Q4	(B)
Q5	(C)
Q6	556~556
Q7	(A)

Hints & Solutions

Q1 Text Solution:

PC value when the given address is in execution $= 240 + 2 = 242$

Target address given = 140

In PC relative mode target address = PC value + relative address

$140 = 242 +$ relative address

Relative address $= 140 - 242 = -102$

Q2 Text Solution:

With in an instruction operation code (opcode), source and destination operand references are specified in general. But next instruction reference is not because for that CPU maintains a register program counter (PC).

Q3 Text Solution:

To access table data (contiguous data) efficiently Auto-increment or Auto-decrement mode are used because using single instruction of these modes, entire table can be accessed without changing the instruction.

Q4 Text Solution:

Mnemonic means the character short form of opcodes, like for addition ADD, for subtraction SUB etc. So the addressing mode in which operand is specified within opcode itself is implied mode.

Q5 Text Solution:

$R1 \leftarrow \#5$, in this instruction source value 5 is mentioned in instruction, hence it is immediate mode

$R1 \leftarrow M[5000]$, in this instruction source value is taken from memory and memory address 5000 is mentioned in instruction hence it is direct mode

$R1 \leftarrow M[R2]$, in this instruction source value is taken from memory and memory address is given indirectly in register R2, hence it is register indirect mode.

Q6 Text Solution:

In PC relative mode target address = PC value + relative address

$720 = PC + 160$

$PC = 720 - 160$

$PC = 560$

PC value will be the address of next instruction when the current instruction is in execution, and current instruction is stored in memory on 4 locations, hence starting address of current instruction $= 560 - 4 = 556$

Q7 Text Solution:

In register mode operand is present in CPU register, hence memory access is not required for operand.

In register indirect mode the register will provide memory address where the operand is stored, hence memory access is required.

In Indirect mode the operand is taken from memory only.

In indexed mode the address of operand is obtained by adding base address in index register value but operand is obtained from memory only.